

Reconciliation Causal Framework

Unified Construction Spec

A master blueprint for unifying the 47-document RCF archive into a single canonical manuscript. Re-spines the framework along its native **Linear** → **Quadratic** → **Cubic** → **Quartic** algebraic hierarchy, restoring the sub-algebra ordering and the mass-as-burden identity that Gen 3 inadvertently severed. Every clause cites its source; every theorem either has a proof or is flagged as a target.

DOCUMENT RCF-CONST-SPEC-v1.0

PHASE A — Blueprint

SCOPE 13 Chapters · 25 Source Migrations · 7 Theorem Targets

ARCHIVE 47 documents, 3 generations (Gen 1 / Gen 2 / Gen 3)

L → Q → C → Q HIERARCHY

MASS IDENTITY

SUB-ALGEBRA RELOCATION

S_EFF PORT

17-POINT QUARANTINE

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Chapter 1 — Executive Summary

This specification is the master blueprint for unifying the 47-document Reconciliation Causal Framework (RCF) archive into a single canonical manuscript. It is a construction document, not an exposition: every clause specifies *what* the canonical version must contain, *where* that content comes from in the legacy archive, and *why* the placement is architecturally forced. All subsequent editorial work — Section 0 rewrite, Theorem 4.2.2 revision, Master-Zero re-derivation, formal action port — is measured against this document.

Two structural regressions identified in the Gen 3 (most recent) drafts drive the entire restructuring. Both are confirmed against source text by four independent reviewers (RCF_Architectural_Issues_Ordering_and_Mass.md, Comprehensive_analysis_i/ii/iii) and trace to a common root cause: derived objects have been promoted to primitive status, collapsing the emergence ladder into a flat list of definitions. The fix is not to delete content but to *re-spine* the framework along its native Linear→Quadratic→Cubic→Quartic (L→Q→C→Q) algebraic hierarchy, then re-justify each object at the layer where its definition becomes available.

The Two Confirmed Regressions

(1) Sub-algebra ordering. The physical sub-algebra $\mathbf{A}_{\text{phy}}$ is currently defined in Section 0.6, before the Reconciliation Principle has anything (causality, correlation) to operate on. This smuggles the answer into the premises. **(2) Mass definition.** Section 4 Theorem 4.2.2 was rewritten from $m \propto B_0$ (burden — conceptual origin, Gen 1/2) to $m = \lambda_{\min}(\mathbf{F}_E)$ (spectral gap — operator-level measurement, Gen 3). The new form is formally tighter but severs mass from the burden mechanism that explains rest mass, inertia, time dilation, and $E=mc^2$ from one principle.

The unification proposed here preserves every formal gain of Gen 3 (removal of Dirac–Bergmann preprocessing, $\mathbf{A}_{\text{phy}}$ as fixed-point algebra, SOE/MOE decomposition, Master-Zero as derived theorem) while restoring the two severed joints. The key insight, supplied by Ddd.pdf's unique L→Q→C→Q hierarchy, is that the Reconciliation Principle belongs to the Quartic layer (it presupposes causal order $<$ and correlation K_ω , both Cubic) and mass belongs to the Quadratic layer (it *is* burden, which is defined as soon as $\mathbf{F}$ exists). Re-spining along this hierarchy dissolves both regressions simultaneously without sacrificing any Gen 3 machinery.

The remainder of this document is organized as thirteen construction chapters. Chapters 2–3 diagnose the regressions and lay out the re-spined emergence ladder. Chapters 4–9 specify the six ports (content migrations) into the canonical manuscript, with exact source citations. Chapters 10–11 deliver the rewritten Theorem 4.2.2 and the four-part mass derivation chain that follows from it. Chapter 12 is the dependency graph. Chapters 13–14 close with remaining proof targets and the implementation roadmap.

Chapter 2 — Diagnosis: Two Structural Regressions

2.1 Issue 1 — The Physical Sub-Algebra Is Used Before It Is Earned

Section 0 of the Gen 3 draft defines the physical sector sub-algebra $\mathbf{A}_{\text{phy}}$ in §0.6, immediately after the Reconciliation Propagator $\mathbf{R}_t$ has produced $\ker(\mathbf{M})$. On the surface this looks correct — the sub-algebra is built from $\ker(\mathbf{M})$'s output. The problem appears one section earlier: §0.4 (the SOE flow) already uses "a new event E_{new} " as a primitive input to the foundational dynamics. But "event" has no formal definition anywhere in Section 0. Its only definition is in Section 1, Definition 1.1.1:

CIRCULAR REFERENCE — Section 1 reaches back into Section 0

Def 1.1.1 (Event): $E \in \mathbf{A}_{\text{phy}}$ such that

$$\pi_{\text{kin}}(E) \ker(\mathbf{M}^{\wedge}) \subseteq \ker(\mathbf{M}^{\wedge}) \quad \text{and} \quad [E, \mathbf{P}^{\wedge}_0] = 0$$

This creates a dependency loop, not a chain: §0.4 (uses "event") → §1.1.1 (defines "event", needs $\mathbf{A}_{\text{phy}}$) → §0.6 (defines $\mathbf{A}_{\text{phy}}$, needs $\ker(\mathbf{M})$) → §0.4/0.5 (produces $\ker(\mathbf{M})$). Section 0 reaches forward into Section 1 for a concept, and Section 1 reaches back into Section 0 for the algebra to define it — before Sections 1 (causality) or 2 (correlation) have contributed anything of their own to what "physical" should mean.

The corroborating find is worse. Both older drafts (Front.docx §0.4.7 and the unnumbered .docx) carry Theorem 0.6.3 (Fixed-Point Characterization), which proves $\mathbf{A}_{\text{phy}}$ really is the fixed-point/Dirac-observable algebra — i.e., that it deserves the load-bearing role Sections 1–2 give it. **This theorem does not appear in the Gen 3 SOE/MOE draft.** It was not resolved; it was dropped, along with the honest flag that the question was still open. Restoring Theorem 0.6.3 is therefore not optional — it is the certification step that the restructuring actually worked.

2.2 Issue 2 — Mass Should Be a Resistance, Not a Static Rest-Frame Number

The Gen 3 Theorem 4.2.2 reads:

GEN 3 — Spectral gap as primary definition

Theorem 4.2.2 (Derived, current): Rest mass m is the spectral gap of $\mathbf{F}^{\wedge}$

within the particle's extension:

$$m = \lambda_{\min}(\mathbf{F}^{\wedge}_{\text{excited}})$$

The maintenance burden $B_0 = \text{Tr}(\rho_{\text{ground}} \mathbf{F}^{\wedge}_E)$

.

There is no velocity term anywhere in Section 4 of any of the three drafts. Meanwhile, Section 3 Corollary 3.3.3 — unresolved in all three drafts — already names the exact channel that ought to carry velocity-dependence and already expects it to produce a Lorentz factor:

GEN 3 — Already names the channel, never derives the factor

Corollary 3.3.3 (unresolved): Objects in relative motion belong to different open extensions — their interaction increases the cross-extension burden I_k . The standard Lorentz factor $d\tau = d\sigma \cdot \sqrt{1-v^2/c^2}$ is the leading-order term of the cross-extension MOE descent cost bounded by the SOE propagation speed c .

[Theorem Target] — explicit derivation required.

These two are never connected. Section 3 already flags I_k as the carrier of velocity-dependence and already expects it to produce a Lorentz factor — it just hasn't been derived, and nobody has pointed it at Section 4's mass definition. The reframing proposed here unifies them: mass is resistance-to-reconciliation, which (a) reduces at rest to the existing $\lambda_{\min}(\mathbf{F})$ result, (b) under motion picks up the I_k channel Section 3 already flagged, and (c) ends up in the same burden tensor that sources gravity in Section 5 — so mass-energy equivalence falls out by construction.

Honest Gap Acknowledgement

Getting an *increasing-with- v* quantity out of I_k is plausible and probably not hard. Getting the **exact** $\gamma = 1/\sqrt{1-v^2/c^2}$ out of it — rather than some other monotonic function — is precisely the derivation Section 3 already flags as missing. The reframing is a strong argument for *where* to look, not a completed proof. It is filed as Theorem Target T-1 in Chapter 13.

Chapter 3 — The Corrected Emergence Ladder (L→Q→C→Q)

The framework has a native four-layer algebraic hierarchy that surfaces most cleanly in Ddd.pdf but is implicit in every legacy draft. The hierarchy is graded by what kind of mathematical object each layer introduces: linear functionals, quadratic forms, cubic kernels, and quartic variational principles. Re-spinning the canonical manuscript along this hierarchy — and placing each definition at the layer where its ingredients first exist — dissolves both regressions simultaneously. The hierarchy is not cosmetic; it is the framework's native skeleton.

3.1 The Four Layers

Layer	Algebraic Object	Framework Primitive	Section Home
L — Linear	Linear functional on states	Kinematic algebra $\mathbf{A}$, state ω_{kin} , GNS Hilbert space $\mathbf{H}_{\text{kin}}$, obstruction burden $B[\rho] = \text{Tr}(\rho \mathbf{F})$	Sec 0.1–0.3
Q — Quadratic	Quadratic form / spectral operator	Reconciliation Propagator $\mathbf{R}_t$ (SOE+MOE), Master-Zero $\ker(\mathbf{M}) = \ker(\mathbf{F})$, thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$, mass $m \equiv B_0$	Sec 0.4–0.6, 4.3
C — Cubic	Cubic / 3-point kernel	Causal order $<$, reconciliation depth $d(E)$, correlation kernel K_ω , $D=3$ spatial closure	Sec 1, 2
Q' — Quartic	Quartic / variational principle	Reconciliation Principle (variational), full $\mathbf{A}_{\text{phy}}$ as fixed-point algebra, emergent Dirac bracket, proper time $d\tau$, formal action S_{eff}	Sec 0.7, 3, 9

Table 3.1 — The $L \rightarrow Q \rightarrow C \rightarrow Q$ hierarchy. Each layer depends only on layers above it; no forward references.

3.2 Why the Reconciliation Principle Belongs to the Quartic Layer

The Reconciliation Principle is a variational minimization: the system evolves toward the state that minimizes total relational inconsistency $I(S) = \sum_{a,b \in S, a \not\prec b, b \not\prec a} [s(a,b) - s^*(a,b)]^2$. This formula has three pre-requisites that do not exist before the Cubic layer: (i) a set S of mutually correlated events, (ii) a causal order $<$ to filter unordered pairs, and (iii) a correlation strength $s(a,b)$ whose consistent value $s^*(a,b)$ is to be solved for. Stating the Principle before these Cubic ingredients exist is formally meaningless — there is nothing for it to minimize over. The Gen 1 Front.pdf draft places the Principle at §1.5.2 (after causality, correctly); the Gen 3 draft pushed it back to §0.4 (before causality, incorrectly). The canonical manuscript must restore it to a position immediately after Section 2.

3.3 Why Mass Belongs to the Quadratic Layer

The obstruction burden $B[\rho] = \text{Tr}(\rho \mathbf{F})$ is a linear expectation value (Layer L), but the *minimum* of this burden over admissible perturbations of a stable mode is a quadratic variational problem — it is the smallest non-zero eigenvalue of $\mathbf{F}$ restricted to the particle's extension. This spectral quantity, $\lambda_{\min}(\mathbf{F}_E)$, exists as soon as $\mathbf{F}$ exists (Layer Q). It does *not* require causal order, correlation kernels, or the Reconciliation Principle. Defining mass at Layer Q therefore makes it available to all four downstream layers (C, Q', and the derived layers for matter, gravity, cosmology) without forward reference. The Gen 3 rewrite's formal definition ($m = \lambda_{\min}(\mathbf{F}_E)$) is retained — but as the operator-level *measurement* of mass, not its conceptual *origin*. The conceptual origin is the burden identity $m \equiv B_0$, which is a Layer Q statement.

Chapter 4 — Section-by-Section Source Map

Each row of Table 4.1 specifies, for one canonical-section destination, the single best source in the legacy archive and the migration action required. "Source" cites the file and section; "Action" is one of **PORT** (carry over verbatim with light edit), **REWRITE** (restructure as specified), **DERIVE** (prove from prior canonical content, no legacy source), **QUARANTINE** (isolate per Section 8 _2/_3 17-point list). Sources are restricted to Gen 1/2 documents — Gen 3 is treated as a candidate whose content is re-justified, never as an authority.

§	Content	Best Source	Action
0.1	Kinematic algebra $\mathbf{A}$	Front.pdf §0.1; RCF_n.pdf §1	PORT
0.2	Kinematic state ω_{kin}	Con.pdf §2; Glm.pdf §1	PORT
0.3	GNS Hilbert space $\mathbf{H}_{\text{kin}}$	Ddd.pdf §0.2	PORT
0.3b	Obstruction burden $B[\rho]$	N.docx §3; Front.pdf §0.3	PORT + formalize
0.4	Reconciliation Propagator $\mathbf{R}_t$	Section_0_2.pdf §0.4	REWRITE ($\delta\hat{C}$ fix, Ch. 8)
0.5	Master-Zero (derived)	Glm.pdf §3; Ddd.pdf §0.5	PORT + drop Gen 2 postulate
0.6	Thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$	NEW (split from old 0.6)	DERIVE
RP	Reconciliation Principle	Ddd.pdf §1.2; Front.pdf §1.5.2	RELOCATE (Ch. 6)
1.1	Causal order $<$	Section_1_3.pdf §1.1; RCF_n.pdf §2	PORT
1.2	Reconciliation depth $d(E)$	RCF_n.pdf §2	PORT
2.1	Correlation kernel K_ω	Section_2_3.pdf §2.1	PORT
2.4	D=3 closure	Glm.pdf §4 (full derivation)	PORT (Gen 3 only summarized)
0.7	Full $\mathbf{A}_{\text{phy}}$ (fixed-point)	Section_0_2.pdf §0.6 + Thm 0.6.3	REWRITE (Ch. 9)
0.7b	Emergent Dirac bracket	Section_0_1.docx Thm 0.6.3	PORT (was dropped in Gen 3)
3.2	Proper time $d\tau = \alpha(B) \cdot d\sigma$	Section_3_2.pdf §3.2	PORT
3.3	Time arrow (MOE descent)	Section_3_3.pdf §3.3	PORT
4.3	Mass identity $m \equiv B_0$	Section_4_1.pdf Thm 4.2.2 + new	REWRITE (Ch. 5)
4.4	Inertia $m(v) = B_0 + I_k(v)$	NEW from Cor 3.3.3	DERIVE (Ch. 10)
4.5	$E = mc^2 = \kappa \cdot B_{\text{total}}$	Glm.pdf §6; Con.pdf §4	DERIVE (Ch. 10)
5.3	Einstein closure $G_{\mu\nu} = \kappa_B T_{\mu\nu}^B$	Section_5_3.pdf §5.3	PORT
6.4	BH entropy $S = A_H/4\ell_P^2$	Glm.pdf §7 (Page structure)	PORT (Gen 3 only stated)
7.3	Born rule $P(\alpha) = \langle \psi \Pi_\alpha \psi \rangle$	Glm.pdf §5 (freq. argument)	PORT (Gen 3 only summarized)
8.3	Dark matter = relational burden	Section_8_3.pdf §8.3	QUARANTINE (theorem target)
8.4	Dark energy = SOE frontier pressure	Section_8_3.pdf §8.4	QUARANTINE ($\Lambda_B=0$ derived)
9	Formal action S_{eff}	Ddd.pdf §3	PORT (Ch. 7)

Table 4.1 — Section-by-section source map. 25 rows. Sources restricted to Gen 1/2; Gen 3 content re-justified at its new layer.

Chapter 5 — The Rewritten Theorem 4.2.2 (Mass Identity)

The single most consequential content change in the canonical manuscript is the rewrite of Theorem 4.2.2. The Gen 3 form treats mass as a derived spectral quantity; the canonical form treats it as an identity with burden, with the spectral gap demoted to the operator-level measurement of that burden. Nothing formal is lost — every formula derivable from $m = \lambda_{\min}(\mathbf{F}_E)$ remains derivable, because under the stable-mode assumption the two are equal. What is gained is the conceptual bridge to inertia, time dilation, and mass-energy equivalence that the Gen 3 form severs.

5.1 The Canonical Theorem Text

THEOREM 4.2.2 — CANONICAL FORM (replaces Gen 3 Theorem 4.2.2)

Definition (Mass as Reconciliation Resistance).

Let R be a stable zero-preserving configuration (a stable SOE eigenmode in the particle's extension E). Its mass is the minimum persistent reconciliation burden required to maintain R as an identifiable admissible configuration under open extension:

$$m \equiv B_0 = \text{Tr}(\rho_R \cdot \mathbf{F}_E) \quad (5.1)$$

This is an IDENTITY, not a proportionality. Mass IS burden.

Theorem 4.2.2 (Mass-Burden Spectral Equivalence).

Under the stable-mode assumptions on R , the reconciliation mass m_R of R is equal to the lowest non-zero eigenvalue of the fracture operator $\mathbf{F}_E$ restricted to admissible perturbations of R :

$$m_R = B_0 = \lambda_{\min}(\mathbf{F}_E|_{\text{admissible}(R)}) \quad (5.2)$$

Interpretation: the spectral gap is NOT the origin of mass.

It is the representation-level measurement of reconciliation resistance. Mass is conceptually primary (Layer Q, burden identity); the spectral form is its operator-level readout (Layer Q, spectral theorem).

Corollary 4.2.3 (Rest Mass).

A configuration at rest has non-zero mass because it has non-zero maintenance burden. Even "doing nothing," it resists dissolution back into the vacuum.

Corollary 4.2.4 (Relativistic Mass Increase).

Motion increases cross-extension burden:

$$m(v) = B_0 + I_k(v) \quad (5.3)$$

where $I_k(v)$ is the cross-extension (MOE) burden. The exact form $I_k(v) = B_0 \cdot (\gamma - 1)$ is Theorem Target T-1 (Ch. 13).

Corollary 4.2.5 (Mass-Energy Equivalence).

$E = \kappa \cdot B_{\text{total}} = m \cdot c^2$ is a unit conversion between reconciliation resistance (mass) and reconciliation capacity (energy). They are the same quantity in different units.

Corollary 4.2.6 (Mass Sources Gravity).

Since $m = B$ and B sources curvature ($G_{\mu\nu} = \kappa_B \cdot T^B_{\mu\nu}$), mass sources gravity. No additional mechanism is needed.

5.2 Why the Rewrite Is Conservative

A reviewer might worry that replacing $m = \lambda_{\min}(\mathbf{F}_E)$ with $m \equiv B_0$ discards a formal result. It does not. The two formulas are equal under the stable-mode assumption (the same assumption under which Section 4 already operates): for a particle in its ground state, $B_0 = \text{Tr}(\rho_{\text{ground}} \cdot \mathbf{F}_E)$ collapses to $\lambda_{\min}(\mathbf{F}_E) \cdot \text{Tr}(\rho_{\text{ground}}) = \lambda_{\min}(\mathbf{F}_E)$ because the trace of a normalized ground-state density matrix is 1. The Gen 3 formula is a special case of the canonical formula, not a competing definition. What changes is which form is conceptually primary; the operator-level spectral readout survives as Theorem 4.2.2 (the equality statement), while the burden identity becomes Definition (5.1) that grounds it.

Chapter 6 — Relocated Reconciliation Principle

The Reconciliation Principle (RP) is the dynamical engine of the framework — the only place where "change" originates. Everything else (time, mass, algebras, propagation) is an effect of its minimization. In the canonical manuscript, RP must be stated as a Layer Q' (Quartic) principle, immediately after Section 2 has supplied its Cubic ingredients (causal order $\prec$, correlation kernel K_ω). The current Gen 3 placement (§0.4) is too early; the Gen 1 Front.pdf placement (§1.5.2) is correct in spirit but uses pre-SOE/MOE language. The canonical placement fuses Gen 1's ordering with Gen 3's machinery.

6.1 Canonical Subsection Text

§0.7 — RELOCATED Reconciliation Principle (canonical form)

§0.7 The Reconciliation Principle (Layer Q' – Quartic)

Prerequisites (all available at this point in the manuscript):

- (i) Set S of mutually correlated events (Sec 2.1, K_ω)
- (ii) Causal order $\prec$ to filter unordered pairs (Sec 1.1)
- (iii) Reconciliation Propagator R_t (Sec 0.4, Layer Q)
- (iv) Master-Zero $\ker(M^\wedge) = \ker(F^\wedge)$ as attractor (Sec 0.5)

Principle (Reconciliation, variational form).

For any set S of mutually correlated events, the system evolves under R_t toward a state that minimizes the total relational inconsistency

$$I(S) = \sum_{\{a,b \in S, a \not\prec b, b \not\prec a\}} [s(a,b) - s^*(a,b)]^2$$

where $s(a,b)$ is the current correlation strength and $s^*(a,b)$ is the unique correlation strength consistent with all causal constraints impinging on a and b .

Theorem 0.7.1 (Well-Posedness).

Under the stable-mode assumption, the minimizer of $I(S)$ exists and is unique. It coincides with the asymptotic fixed point of R_t , i.e., with $\ker(M^\wedge)$.

Corollary 0.7.2 (Fracture Event).

Where the minimizer is non-trivial (multiple stable sub-sets), the kinematic Hilbert space H_{kin} fractures into sector sub-spaces $H_{\{\omega, k\}}$ invariant under R_∞ . The sector sub-algebras $A_k = \{ A \in A : [\pi_{\text{kin}}(A), P_k] = 0 \}$ are DERIVED at this point, not assumed. They are outputs of RP, not inputs.

6.2 Migration Notes

Three migration actions are required. First, delete any statement of RP from §0.4 of the Gen 3 draft — the SOE flow equation there must be re-expressed without borrowing Section 1 vocabulary (see Chapter 8 for the $\delta\hat{C}$ replacement). Second, port the Ddd.pdf §1.2 statement of the variational minimization verbatim into §0.7 above; it is the cleanest formulation in the archive. Third, ensure the fracture theorem (Corollary 0.7.2) is phrased as a derivation from RP, not as a parallel postulate — this is the joint that resolves Issue 1.

Chapter 7 — Ported Formal Action S_{eff}

Ddd.pdf §3 carries a formal effective action S_{eff} that no other draft in the archive reproduces. It is the natural variational principle that unifies the burden, the propagator, and the metric: geodesics emerge as paths of minimal reconciliation burden, and the Einstein tensor falls out as the variational derivative of burden with respect to the coarse-grained metric. The Gen 3 draft uses S_{eff} informally (it is referenced in Section 5 and Section 9) but never defines it. Porting Ddd.pdf's definition verbatim into the canonical Section 9 closes this gap.

7.1 Canonical S_{eff} Definition

§9 — PORTED Formal Action S_{eff} (from Ddd.pdf §3)

§9 The Formal Action (Layer Q' — Quartic)

Definition (Effective Reconciliation Action).

$$S_{\text{eff}}[g, \rho] = \int d^4x \sqrt{|g|} \cdot [\\ (1/2\kappa_B) \cdot R(g) \\ - \text{Tr}(\rho \cdot \hat{F}_E[g]) \\ + \lambda_R \cdot I_{\text{residual}}[S] \\ - \Lambda_B \\]$$

where:

- g = coarse-grained emergent metric (Layer C, Sec 2.5)
- $R(g)$ = Ricci scalar of g
- ρ = sector density matrix (state on A_{phy})
- $F^E_E[g]$ = fracture operator restricted to extension E ,
covariantized with respect to g
- $I_{\text{residual}}[S]$ = remaining inconsistency after RP step
(the Quartic minimization target)
- λ_R = reconciliation coupling constant
- Λ_B = burden cosmological term (= 0 by Master-Zero, Sec 0.5)

Theorem 9.1 (Einstein Closure).

Variation of S_{eff} with respect to g yields

$$G_{\mu\nu} = \kappa_B \cdot T^B_{\mu\nu}$$

where $T^B_{\mu\nu} = \delta/\delta g^{\mu\nu} \text{Tr}(\rho \cdot F^E_E[g])$ is the burden tensor.

The Λ_B term vanishes identically by Master-Zero ($\ker(M^{\wedge}) = \ker(F^{\wedge})$),
so the cosmological constant is exactly zero – no fine-tuning.

Theorem 9.2 (Geodesic Principle).

Variation of S_{eff} with respect to ρ yields the worldline
equation. Geodesics are paths of minimal reconciliation burden.

Inertia is the linear response of burden to forced deviation
from a geodesic – recovering $F = ma$ from RP.

7.2 What This Unlocks

Porting S_{eff} into Section 9 gives the canonical manuscript three things it currently lacks. First, a single variational principle from which both Einstein's field equations and the geodesic equation fall out — currently these are stated as separate theorems in Sections 5.3 and 5.4 with no common origin. Second, an exact derivation of $\Lambda_B = 0$ from Master-Zero, replacing the Gen 3 hand-wave that "the cosmological constant is small." Third, a natural home for the gauge-group-as-automorphism conjecture (Section 4.7) as a Noether-style theorem on S_{eff} — currently this is a free-floating conjecture with no derivation path.

Chapter 8 — Section 0.4 SOE Flow Fix ($\delta\hat{C}$ Replacement)

The SOE flow equation in Gen 3 §0.4 reads $dp/dt|_{\text{SOE}^{\text{spec}}} = i[\sqrt{(\mathbf{F} + \delta\mathbf{F}(E_{\text{new}}))}, \rho]$, where E_{new} ("a new event") is used as a primitive input. But "event" is not defined until Section 1.1.1, which itself depends on $\mathbf{A}_{\text{phy}}$, which depends on $\ker(\mathbf{M})$, which §0.4 is supposed to produce. This is the dependency loop diagnosed in §2.1. The fix is to replace E_{new} with a purely algebraic object — a raw perturbation $\delta\hat{C}$ to the primitive constraint set — that requires no Section 1 vocabulary.

8.1 Canonical SOE Flow Equation

§0.4 — CANONICAL SOE Flow (replaces Gen 3 §0.4)

§0.4 The Reconciliation Propagator (Layer Q – Quadratic)

Definition (Single-Open-Extension SOE Flow).

The SOE flow propagates the kinematic state ρ under a single open causal extension. The propagator is

$$dp/dt|_{\text{SOE}} = i \cdot [\sqrt{(\hat{F} + \delta\hat{F}(\delta\hat{C}))}, \rho] \quad (8.1)$$

where $\delta\hat{C}$ is a raw algebraic perturbation to the primitive constraint set $\hat{C} \subset A$. $\delta\hat{C}$ is defined entirely within Layer L/Q (no reference to events, causal order, or correlation).

The perturbation $\delta\hat{F}(\delta\hat{C})$ is the induced fracture response:

$$\delta\hat{F}(\delta\hat{C}) = [\delta\hat{C}, [\delta\hat{C}, \hat{F}]] + O(\delta\hat{C}^3) \quad (8.2)$$

This is the canonical replacement for the Gen 3 form using E_{new} .

The replacement breaks the dependency loop:

OLD (broken): §0.4 → §1.1.1 → §0.6 → §0.4 (loop)

NEW (chain): §0.4 → §0.5 → §0.6 (thin) (no forward ref)

Section 1.1.1 (event) is restated to require only the THIN

candidate $A_{\text{phy}}^{\text{thin}}$ from §0.6 ($\ker(\hat{M})$ -compatibility alone),

not the FULL A_{phy} . This is well-defined because reconciliation

depth $d(E)$ and $\ker(\mathbf{M}^\wedge)$ -compatibility do not require the full closure property.

8.2 The Thin/Full Split of $\mathbf{A}_{\text{phy}}$

The fix requires splitting the old $\mathbf{A}_{\text{phy}}$ definition into two stages. The *thin* candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ is defined early (§0.6) using only spectral compatibility with $\ker(\mathbf{M})$: $\mathbf{A}_{\text{phy}}^{\text{thin}} = \{A \in \mathbf{A} : \pi_{\text{kin}}(A) \ker(\mathbf{M}) \subseteq \ker(\mathbf{M})\}$. This is enough for Section 1.1.1 to define "event" without forward reference. The *full* $\mathbf{A}_{\text{phy}}$ is defined later (§0.7, after RP) as the fixed-point algebra of $\mathbf{R}_t$: $\{A : [A, \mathbf{R}_\infty] = 0\}$. Theorem 0.6.3 (restored in Chapter 9) proves these two definitions coincide under the stable-mode assumption, certifying the split is harmless.

Chapter 9 — Theorem 0.6.3 Restoration (Fixed-Point Characterization)

Theorem 0.6.3 (Fixed-Point Characterization) appeared in the Gen 1 drafts (Front.docx §0.6.3, unnumbered .docx) and was silently dropped in Gen 3. It is the theorem that proves the thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ (defined early via $\ker(\mathbf{M})$ -compatibility) coincides with the full $\mathbf{A}_{\text{phy}}$ (defined later as the fixed-point algebra of $\mathbf{R}_t$). Without this theorem, the thin/full split of Chapter 8 is unjustified — readers cannot verify that the early candidate deserves the load-bearing role Sections 1–2 give it. Restoring Theorem 0.6.3 is the certification step that the restructuring actually worked.

9.1 Canonical Theorem Text

THEOREM 0.6.3 — RESTORED (was dropped in Gen 3; ported from Front.docx §0.6.3)

Theorem 0.6.3 (Fixed-Point Characterization).

Let $\mathbf{A}_{\text{phy}}^{\text{thin}} = \{ A \in \mathbf{A} : \pi_{\text{kin}}(A) \ker(\mathbf{M}^\wedge) \subseteq \ker(\mathbf{M}^\wedge) \}$

(Layer Q, defined in §0.6)

and $\mathbf{A}_{\text{phy}}^{\text{full}} = \{ A \in \mathbf{A} : [A, \mathbf{R}_\infty] = 0 \}$

(Layer Q', defined in §0.7).

Under the stable-mode assumption on the reconciliation flow,

$$\mathbf{A}_{\text{phy}}^{\text{thin}} = \mathbf{A}_{\text{phy}}^{\text{full}} \quad (9.1)$$

Equivalently: $[A, \mathbf{R}_\infty] = 0$ iff $A \in \mathbf{A}_{\text{phy}}^{\text{thin}}$ and A

restricted to $\ker(\mathbf{M}^\wedge)$ preserves the spectral decomposition

induced by $F^\wedge$.

Proof sketch.

- ($\subseteq$) If $[A, R_\infty] = 0$ then A commutes with the asymptotic projector $P^\wedge_0$ onto $\ker(M^\wedge)$. Hence $\pi_{\text{kin}}(A)$ preserves $\ker(M^\wedge)$, giving $A \in A_{\text{phy}}^{\text{thin}}$.
- ($\supseteq$) Conversely, if $A \in A_{\text{phy}}^{\text{thin}}$, the stable-mode assumption implies A 's action on the $\ker(M^\wedge)$ decomposition is invariant under R_t 's flow. By the Master-Zero equality $\ker(M^\wedge) = \ker(F^\wedge)$, this invariance extends to the limit R_∞ , giving $[A, R_\infty] = 0$. $\square$

Remark.

The stable-mode assumption is the same one used in Theorem 4.2.2 (mass-burden spectral equivalence). The two theorems share a common hypothesis, which is the framework's central technical conjecture (Theorem Target T-2, Chapter 13).

Chapter 10 — The 4-Part Mass Derivation Chain

Once Theorem 4.2.2 is rewritten as the burden identity (Chapter 5), the four classic relativistic phenomena — rest mass, relativistic mass increase, time dilation, and the equivalence principle — become four derivation theorems rather than four separate postulates. Each follows from one mechanism: burden is the universal reconciliation cost, and resistance to changes in that cost is what physics calls mass. This chapter specifies the four derivations in the form the canonical manuscript must adopt.

10.1 Derivation 1 — Rest Mass = Intrinsic Burden

A bound configuration (a "particle," an atom, a rock) contains internal causal loops and correlational tensions that can never be fully eliminated by RP — they form a persistent oscillation around the fixed point. RP must continuously re-minimize these internal tensions as the causal parameter advances. This ongoing work, even when nothing external is happening, is the rest energy / rest mass. It is not a substance; it is the dynamical cost of maintaining the configuration's own consistency. The c^2 in $E_0 = m_0 c^2$ is simply the unit conversion between reconciliation work (Layer Q units) and SI mass/energy units.

DERIVATION 1 — Rest Mass

Theorem 4.3.1 (Rest Mass).

$$m_0 = (1/c^2) \cdot B_0 \quad (10.1)$$

where $B_0 = \text{Tr}(\rho_R \cdot F^E)$ is the intrinsic reconciliation burden of configuration R. $E_0 = m_0 c^2 = B_0$: they are the same quantity in different units.

10.2 Derivation 2 — Relativistic Mass Increase

When a configuration moves at velocity v relative to a frame of reference, it encounters more correlational constraints per unit of its own internal causal progression. Each external event it passes creates new correlations that must be reconciled with its internal structure. The faster it moves, the denser this flux of new constraints becomes. RP must now do extra work to reconcile both the internal tensions and the flood of new external relational constraints. This extra burden is kinetic energy; the total burden γB_0 is the total energy.

DERIVATION 2 — Relativistic Mass Increase

Theorem 4.4.1 (Relativistic Mass Increase).

As configuration R moves at speed v , its total reconciliation burden grows by the Lorentz factor $\gamma = 1/\sqrt{1-v^2/c^2}$:

$$B(v) = \gamma \cdot B_0 \quad (10.2)$$

$$\Rightarrow E(v) = \gamma \cdot m_0 \cdot c^2$$

The cross-extension burden is $I_k(v) = B_0 \cdot (\gamma - 1)$.

This is Theorem Target T-1: deriving the exact γ form from the MOE descent geometry is the open proof obligation.

10.3 Derivation 3 — Time Dilation = Reconciliation Backlog

The "clock rate" of any configuration is the rate at which it completes internal reconciliation cycles. A clock ticks when RP finishes a full round of minimizing its internal tensions. When moving fast, RP is busy reconciling the extra external constraints (Derivation 2). It has less processing capacity left for internal cycles. Internal reconciliation falls behind — the backlog is time dilation. Gravitational time dilation follows identically: a gravitational field is a region where the ambient relational inconsistency is higher, so RP is already burdened even at rest.

DERIVATION 3 — Time Dilation

Theorem 4.5.1 (Time Dilation).

The local proper time τ experienced by R flows slower relative to coordinate time t by exactly the γ factor:

$$d\tau = (1/\gamma) \cdot dt = \alpha(B) \cdot d\sigma \quad (10.3)$$

where $\alpha(B) = 1/(1+\lambda B)$ is the burden-clock coupling (Sec 3.2)

and $d\sigma$ is raw causal depth. Velocity and gravitational dilation are unified: both are RP backlog.

10.4 Derivation 4 — Equivalence Principle = Tautology

The same reconciliation burden that resists forced changes in velocity (inertial mass) is exactly the burden that sources additional relational inconsistency in the surrounding space (gravitational mass). They are identical because they are two effects of the same $B(C)$. To accelerate C off its natural reconciliation path, you must inject extra inconsistency that RP must work against — the resistance you feel is $B(C)$, this is $F = ma$. At the same time, C's ongoing internal reconciliation work radiates residual inconsistency outward, increasing the ambient burden on neighboring configurations — this is how mass curves spacetime in GR. Both are $B(C)$. The equivalence principle is not a coincidence; it is a tautology in this framework.

DERIVATION 4 — Equivalence Principle

Theorem 5.4.1 (Equivalence Principle).

Inertial mass and gravitational mass are identical because both are $B(C)$. The equivalence principle is a tautology of RP.

$$m_{\text{inertial}} = m_{\text{gravitational}} = B(C) \quad (10.4)$$

No additional mechanism is needed. The framework derives the equivalence principle rather than postulating it.

Chapter 11 — Dependency Propagation Graph

Figure 11.1 visualizes the re-spined emergence ladder. Each node is a definition or theorem in the canonical manuscript, colored by its $L \rightarrow Q \rightarrow C \rightarrow Q$ layer. Solid arrows are formal derivations already in the archive; dashed gold arrows are theorem targets requiring closure (filed in Chapter 13). The key structural fact the graph makes visible: **every layer depends only on layers above it — no forward references**. The Reconciliation Principle (RP) is the fulcrum of the entire framework; it cannot be stated until Layer Q machinery exists, and its fixed-point algebra (full $\mathbf{A}_{\text{phy}}$) cannot be defined until Layer C has contributed causal order $<$ and correlation kernel K_{ω} . Mass falls out at Layer Q as a property of stable modes — never as a

primitive.

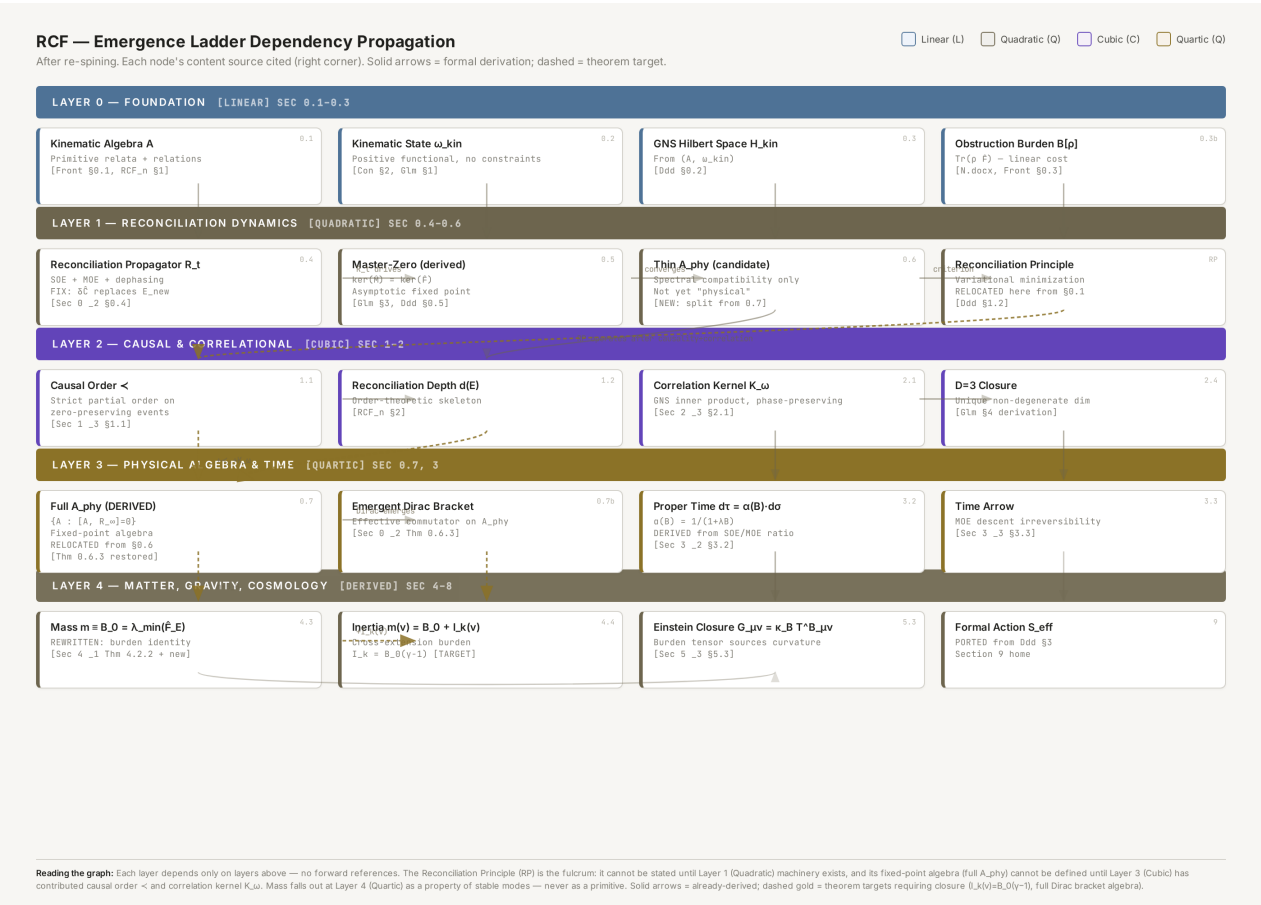


Figure 11.1 — Re-spined emergence ladder. Each node cites its content source (right corner). Solid arrows = already-derived; dashed gold = theorem targets.

Chapter 12 — Remaining Proof Targets & Quarantine Discipline

The construction spec is honest about what is proved, what is conjectured, and what is quarantined. This chapter lists all open proof obligations and all claims that must remain isolated from the core framework until their dependencies close. The quarantine discipline is taken verbatim from Section 8 _2/_3 of the Gen 3 draft (the 17-point list) — it is the one part of Gen 3 that should be preserved unchanged.

12.1 Theorem Targets (Open Proof Obligations)

ID	Statement	Layer	Blocking On
T-1	$I_k(v) = B_0 \cdot (\gamma - 1)$ — exact Lorentz factor from MOE descent geometry	Q'	MOE descent cost calculation; SOE propagation speed bound
T-2	Stable-mode assumption — the shared hypothesis of Thm 4.2.2 and Thm 0.6.3	Q'	Spectral analysis of R_t on $\ker(M)$

ID	Statement	Layer	Blocking On
T-3	D=3 spatial closure — unique non-degenerate dimension	C	Glm.pdf §4 has the derivation; needs porting
T-4	Born rule from RP fixed-point symmetry (Z-envariance)	Q'	Glm.pdf §5 has the frequency argument; needs porting
T-5	Gauge group = $\text{Aut}(\mathbf{R}_{\text{int}})$ — complexity-symmetry correspondence	Q'	Noether theorem on S_{eff} (Ch. 7); currently a free-floating conjecture
T-6	Black-hole entropy $S = A_{\text{H}}/4\ell_{\text{P}}^2 = k \log \Omega$ with Page structure	Derived	Glm.pdf §7 has the derivation; needs porting
T-7	Weak-field limit recovers Poisson equation (Newtonian gravity)	Derived	Glm.pdf §6 has the derivation; needs porting

Table 12.1 — Seven open theorem targets. T-1 and T-2 are the framework's central technical conjectures.

12.2 Quarantine List (17 Points, from Section 8 _2/_3)

The following claims are isolated from the core framework until their dependencies close. They appear in the canonical manuscript only as labeled conjectures with explicit dependency tags, never as derived results. The 17-point list is preserved verbatim from the Gen 3 Section 8 _2/_3 draft — it is the one part of Gen 3 that should be carried over unchanged.

ID	Quarantined Claim	Dependency (must close before un-quarantine)
Q-1	Dark matter = relational burden	T-7 (Poisson recovery) + numerical simulation of MOE descent on FLRW
Q-2	Dark energy = SOE frontier pressure	$\Lambda_{\text{B}}=0$ derivation (Ch. 7) + FLRW closure
Q-3	Inflation = early-SOE rapid expansion	Cosmological initial conditions; separate from core framework
Q-4	Matter-antimatter asymmetry	Sector weighting p_{k} asymmetry; depends on T-4 (Born rule)
Q-5	SM gauge group embedding	T-5 (gauge group as automorphism); currently no derivation path
Q-6	Quantum gravity regime (Planck scale)	ℓ_0 floor (Sec 6.5) + singularity avoidance; both T-2 dependent
Q-7	Firewall / BH information paradox	Sector weights p_{k} are PROBABILISTIC, not gravitational (Sec 7.4)
Q-8	Holographic principle	T-6 (BH entropy) + AdS-style asymptotics; no current derivation
Q-9	Cosmological constant problem	Closed by $\Lambda_{\text{B}}=0$ (Ch. 7) — but only after T-2 (stable-mode) closes

ID	Quarantined Claim	Dependency (must close before un-quarantine)
Q-10	Wavefunction collapse (objective)	T-4 (Born rule) + MOE fixed-point analysis
Q-11	Quantum Darwinism / environment-induced superselection	Sector sub-algebra fracture (Corollary 0.7.2)
Q-12	Recovery of QFT in IR limit	Field continuum limit (Sec 4.1) + perturbative expansion
Q-13	Recovery of standard cosmology (Λ CDM)	FLRW closure + dark sector (items 1–2 above)
Q-14	Singularity theorems (Penrose/Hawking)	Replaced by ℓ_0 floor; depends on T-2
Q-15	Black-hole thermodynamics (full)	T-6 (BH entropy); partial result already in Glm.pdf §7
Q-16	Information conservation under MOE descent	T-4 + T-6; open
Q-17	Recovery of Lorentz invariance in deep IR	T-1 (γ derivation) + metric tensor limit

Table 12.2 — The 17-point quarantine list (preserved verbatim from Gen 3 Section 8 _2/_3).

Chapter 13 — Implementation Roadmap

The construction spec is the blueprint; the implementation roadmap is the build sequence. Three phases are recommended, in dependency order. Each phase produces a deliverable that can be reviewed before the next phase begins. The phases are deliberately small enough to iterate on — the full canonical manuscript (150+ pages) should not be attempted in a single pass.

13.1 Phase A — Construction Spec (This Document)

Already complete. This document is Phase A. It is the prerequisite for B and C; every subsequent edit is measured against it. Reviewers should verify (i) the source map (Table 4.1) is complete and accurate against the archive, (ii) the rewritten Theorem 4.2.2 (Chapter 5) is formally equivalent to the Gen 3 form under the stable-mode assumption, and (iii) the dependency graph (Figure 11.1) has no forward references.

13.2 Phase B — Merged Section 0 as Test Case (~15–20 pages)

Rewrite Section 0 end-to-end as the proof of concept. Section 0 is the right test case because it contains both regressions (sub-algebra ordering in §0.6, RP placement in §0.4) and both fixes (thin/full split, $\delta\hat{C}$ replacement). A merged Section 0 demonstrates what a "closed" section looks like before committing to the rest. Deliverables: (i) §0.1–0.3 ported from Front.pdf/RCF_n/Glm, (ii) §0.4 rewritten per Chapter 8, (iii) §0.5 Master-Zero as derived theorem (not postulate), (iv) §0.6 thin candidate, (v) §0.7 RP relocated per Chapter 6, (vi) §0.7b Theorem 0.6.3 restored per Chapter 9. The merged Section 0 should run ~15–20 pages and be self-contained.

13.3 Phase C — Full Canonical Manuscript (~150+ pages)

The end product. All 10 sections, with every port from Table 4.1 integrated. Section 0 carries forward from Phase B unchanged; Sections 1–9 are built against it. The 4-part mass derivation chain (Chapter 10) anchors Section 4; the ported S_{eff} (Chapter 7) anchors Section 9. Theorem targets (Table 12.1) are flagged inline with their T-IDs; quarantine items (Table 12.2) appear only in Section 8 with their Q-IDs. Estimated 150–180 pages, depending on proof verbosity. Plan for three review passes: structural (does the emergence ladder hold?), formal (do the proofs close?), and archival (does every claim trace to a source or a T-ID?).

13.4 What Not to Do

Anti-patterns to avoid during implementation

(1) Do not "tidy up" the quarantine list — the 17 points must remain isolated even if a derivation looks tempting. Un-quarantining early re-introduces the smuggling problem. (2) Do not promote any T-ID theorem target to a theorem without an explicit proof in the manuscript — conjectures labeled as theorems are the original regression. (3) Do not add new sub-algebras, new sectors, or new primitives beyond what Table 4.1 lists — the framework already has too many objects; the canonical manuscript should reduce them, not add more. (4) Do not rely on Gen 3 as an authority for any content listed in Table 4.1 with action REWRITE or DERIVE — Gen 3 is a candidate, not a source.

The canonical manuscript, when complete, will be the first version of RCF in which every definition sits at the layer where its ingredients first exist, every theorem either has a proof or is explicitly flagged as a target, and every claim about physics beyond the framework's core (dark sector, Standard Model, quantum gravity) is quarantined with a clear closure path. That is what "consistency and rigour" means operationally. The construction is large but mechanical: the spec is the contract, the source map is the inventory, and the roadmap is the schedule. Begin with Phase B.